MILITARY SPECIFICATION

TEST HARNESS, RADIO RECEIVING SET AN/ARM-44(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall hold, position, electrically and mechanically connect, and protect from electrical and mechanical damage Radio Receiving Set AN/ARR-58 during operation and maintenance in a bench test setup.

1.2 Classification - The Test Harness shall consist of the following items:

| Item | Type Designation | Appl. Para. |
|---|---|---|
| Electrical Equipment Maintenance Fixture | MT-2162(*)/ARM-44 | 3.4.1 |
| Electrical Equipment Rack | MT-2163(*)/ARM-44 | 3.4.2 |
| Mounting | MT-2164(*)/ARM-44 | 3.4.3 |
| Band-Pass Filter | F-429(*)/ARM-44 | 3.4.4 |
| Interconnecting Box | J-952(*)/ARR-58 | 3.4.5 |
| Cable Group | | 3.4.6 |
| Electrical Power Cable Assembly | W-1 of AN/ARM-44 | |
| Electrical Power Cable Assembly | W-2 of AN/ARM-44 | |
| Special Purpose Electrical Cable Assembly | W-3 of AN/ARM-44 | |
| Special Purpose Electrical Cable Assembly | W-4 of AN/ARM-44 | |
| Radio Frequency Cable Assembly | W-5 of AN/ARM-44 | |
| Radio Frequency Cable Assembly | W-6 of AN/ARM-44 | |
| Special Purpose Electrical Cable Assembly | W-7 of AN/ARM-44 | |
| Special Purpose Electrical Cable | W-8 of AN/ARM-44 | |
| Special Purpose Electrical Cable Assembly | W-9 of AN/ARM-44 | |
| Special Purpose Electrical Cable Assembly | W-10 of AN/ARM-44 | |
| Special Purpose Electrical Cable Assembly | W-11 of AN/ARM-44 | |
| Radio Frequency Cable Assembly | W-12 of AN/ARM-44 | |
| Radio Frequency Cable Assembly | W-13 of AN/ARM-44 | |

FSC 4940



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## 2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of the invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-T-5422 | Testing, Environmental, Aircraft Electronic Equipment |
| MIL-I-6181 | Interference Control Requirements, Aircraft Equipment |
| MIL-H-15362 | Harnesses; Test Bench, Avionic Equipment, General Specification for |
| MIL-E-17555 | Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of |
| MIL-T-18303 | Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for |
| MIL-N-18307 | Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment |
| MIL-T-21200 | Test Equipment For Use With Electronic and Fire Control Systems, General Specification for |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-242 | Electronic Equipment Parts |

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

## 3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc., set forth in MIL-STD-242.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.



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(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.2 Design and Construction

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 95 pounds.

3.2.3 Operating Life

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 1000 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during normal use.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment shall have a minimum of 1000 hours operating time between failures. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

3.2.4 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-H-15362.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

| | |
|---|---|
| Temperature | Room ambient (30° ±10°C) |
| Altitude | Normal ground |
| Vibration | None |
| Humidity | Room ambient up to 90% relative humidity |
| Input power voltage | 115 ±1.0 V AC, 380-420 CPS, 3-PHASE, and 28 V DC |



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3.2.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-H-15362.

3.2.10 Primary Input Power Requirements - This equipment has no appreciable input power requirements of its own, but shall serve to connect the primary input power to the equipment under test.

3.2.10.1 Voltage Protection - The equipment shall not be damaged by voltages either above or below the steady state and transient voltages specified in MIL-STD-704 and shall automatically resume normal operation when the voltage returns within limits.

3.2.11 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3 Performance - The Test Harness shall be suitable for use on a Shore Shop Bench Arrangement consisting of a Type I Workbench, a Type II Workbench, and an Auxiliary Top, Bureau of Ships No. 1385763.

3.4 Detail Requirements

3.4.1 Electrical Equipment Maintenance Fixture MT-2162/ARM-44

3.4.1.1 Function - The Electrical Equipment Maintenance Fixture MT-2162/ARM-44 shall consist of one set of bench top fixtures to accommodate the high maintenance items of Radio Receiving Set AN/ARR-58.

3.4.1.2 Form Factor - The maximum overall dimensions of MT-2162/ARM-44 shall be 11 inches wide by 29 inches in length by 13 inches in height.

3.4.1.3 Weight - The weight of MT-2162/ARM-44 shall not exceed 19 pounds.

3.4.1.4 Contents - The MT-2162/ARM-44 shall consist of the following:

(1) One trunnion cradle which will accommodate Radio Receiver R-863/ARR-58, and Power Supply PP-1923/ARR-58. The cradle shall rotate through 180° with R-863/ARR-58 in position, and through 320° with PP-1923/ARR-58 in position, in a single plane and shall lock in maintenance position.

(2) One adapter plate which will accommodate Audio Frequency Amplifier AM-1870/ARR-58 or Radio Frequency Amplifier AM-1871/ARR-58 and will lock in maintenance position.

3.4.2 Electrical Equipment Rack MT-2163/ARM-44

3.4.2.1 Function - The Electrical Equipment Rack MT-2163/ARM-44 shall provide mounting for four Radio Receivers R-863/ARR-58 and Power Supply PP-1923/ARR-58 when installed under the Auxiliary Top and Type II Workbench.

3.4.2.2 Form Factor - The maximum overall dimensions of MT-2163/ARM-44 shall be 18 inches wide by 38 inches in length by 20 inches in height.



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3.4.2.3 Weight - The weight of the MT-2163/ARM-44 shall not exceed 66 pounds.

3.4.2.4 Contents - The MT-2163/ARM-44 shall consist of racks, rack slides, mounting plates, shelf plates, and attaching hardware. Clamps and brackets shall secure the equipment under test in fixed position, and position the equipment in such a manner as to allow normal operation and provide for simple and quick attachment of the equipment.

3.4.3 Mounting MT-2164/ARM-44

3.4.3.1 Function - Mounting MT-2164/ARM-44 shall be attached beneath the top shelf of the workbench and shall provide mounting for Monitor Panel MX-2408/ARR-58, Radio Set Control C-2410/ARR-58, and Radio Set Control C-2411/ARR-58.

3.4.3.2 Form Factor - The maximum overall dimensions of Mounting MT-2164/ARM-44 shall be 5 inches wide by 17-1/2 inches in length by 4-1/2 inches in height.

3.4.3.3 Weight - The weight of the MT-2164/ARM-44 shall not exceed 1-3/4 pounds.

3.4.4 Band-Pass Filter F-429/ARM-44

3.4.4.1 Function - Band-Pass Filter F-429/ARM-44 shall perform the audio noise and sensitivity tests for the AN/ARR-58 system under test. The F-429/ARM-44 shall be of the LC type with an input and output impedance of approximately 2500 ohms and shall operate over a frequency range from 0 to 500 cps in four switch-selected bands.

3.4.4.2 Form Factor - The maximum overall dimensions of Band-Pass Filter F-429/ARM-44 shall be 6-1/2 inches in width by 9 inches in length by 6-1/2 inches in height.

3.4.4.3 Weight - The weight of the F-429/ARM-44 shall not exceed 4-1/2 pounds.

3.4.4.4 Electrical Characteristics - The four filter bands covering the frequency range shall be as follows:

Band 1 - Low Pass filter 0 to 200 cps
Band 2 - Band Pass filter 100 to 300 cps
Band 3 - Band Pass filter 300 to 500 cps
Band 4 - Low Pass filter 0 to 500 cps

All four bands shall have a mid-band attenuation of 6 db ± 0.5 db.

Bandpass characteristics shall be as follows:

Band 1 - The frequency response shall be flat within +2 -3 db from 20 to 180 cps with respect to 100 cps, and shall be down no more than 4.5 db at the 200 cps cut off frequency. The response shall be down at least 20 db at 420 cps.

Band 2 - The frequency response shall be flat within +2 -3 db from 120 to 280 cps with respect to 200 cps, and shall be down no more than 4.5 db at the 100 and 300 cps cut off frequencies. The response shall be down at least 20 db at 50 and 520 cps.

Band 3 - The frequency response shall be flat within +2 -3 db from 320 to 480 cps with respect to 400 cps and shall be down no more than 4.5 db at the 300 and 500 cps cut off frequencies. The response shall be down at least 20 db at 220 and 680 cps.

Band 4 - The frequency response shall be flat within +2 -3 db from 20 to 480 cps with respect to 250 cps and shall be down no more than 4.5 db at the 500 cps cut off frequency. The response shall be down at least 20 db at 650 cps.



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3.4.4.5 Controls - A rotary switch shall be provided on the front panel for selecting the desired filter band, and input and output terminals shall be provided.

3.4.5 Interconnecting Box J-952/ARR-58

3.4.5.1 Function - Interconnecting Box J-952/ARR-58 shall provide for electrically and mechanically joining cable assemblies of the test harness and distribute power and audio signals to the AN/ARR-58 system under test. The J-952/ARR-58 shall be the same as that used in the aircraft installation of Radio Receiving Set AN/ARR-58.

3.4.5.2 Form Factor - The maximum overall dimensions of Interconnecting Box J-952/ARR-58 shall not exceed 7 inches in width by 7-3/4 inches in length by 5 inches in height.

3.4.5.3 Weight - The weight of the J-952/ARR-58 shall not exceed 5 pounds.

3.4.5.4 Contents of Unit - The J-952/ARR-58 shall contain connectors for primary power, a three-phase circuit breaker, and four switching relays consisting of two power switching relays and two audio switching relays. The switching circuits shall provide the functions shown in Figure 1. The 115 vac, 400 cps, and the 28 vdc power supplies shall enter the Interconnecting Box through connectors J1014 and J1015, the 28 vdc circuit being protected by a fuse. Phase 1 of the 115 vac supply is applied to Power Supply PP-1923/ARR-58 through relay K1003. Phase 2 and 3 are applied through relay K1004 to the Interconnecting Box Power Supply J-953/ARR-58. Channels B-D and A-C, respectively. Relays K1003 and K1004 shall be controlled by PWR-OFF switch in Radio Set Control C-2411/ARR-58. Relay K1001 shall apply the output from Audio Amplifier A or C to Monitor Panel MX-2408/ARR-58. The audio channel selector, S901, on the Monitor Panel shall be in the A or C position, whichever is desired. Relay K1002 shall apply the output from Audio Amplifier B or D to the Monitor Panel. Switch S901 in the Monitor Panel must be in B or D position.

3.4.6 Cable Group (MGX 708382-1)

3.4.6.1 Function of Interconnecting Cable Assemblies - The interconnecting cables shall provide for electrical connection of units to allow normal operation of the AN/ARR-58 Radio Receiving Set.

3.4.6.2 Quantity and Types of Interconnecting Cable Assemblies - The Test Harness shall include the following Interconnecting Cable Assemblies: (All wire to meet requirements of Specification MIL-W-16878 Type B).

(1) Primary Power Cable W-1. This Cable Assembly shall be approximately 3 feet in length and shall have eight AWG #18 conductors. Three of the conductors shall be shielded. The shop power end of the cable shall have pigtail leads clearly identified. The equipment end of the cable shall have the following:

| P1014 | P1015 |
|---|---|
| MS3106A-22-4S | MS3106A-22-4SX |
| MS3057-12A | MS3057-12A |



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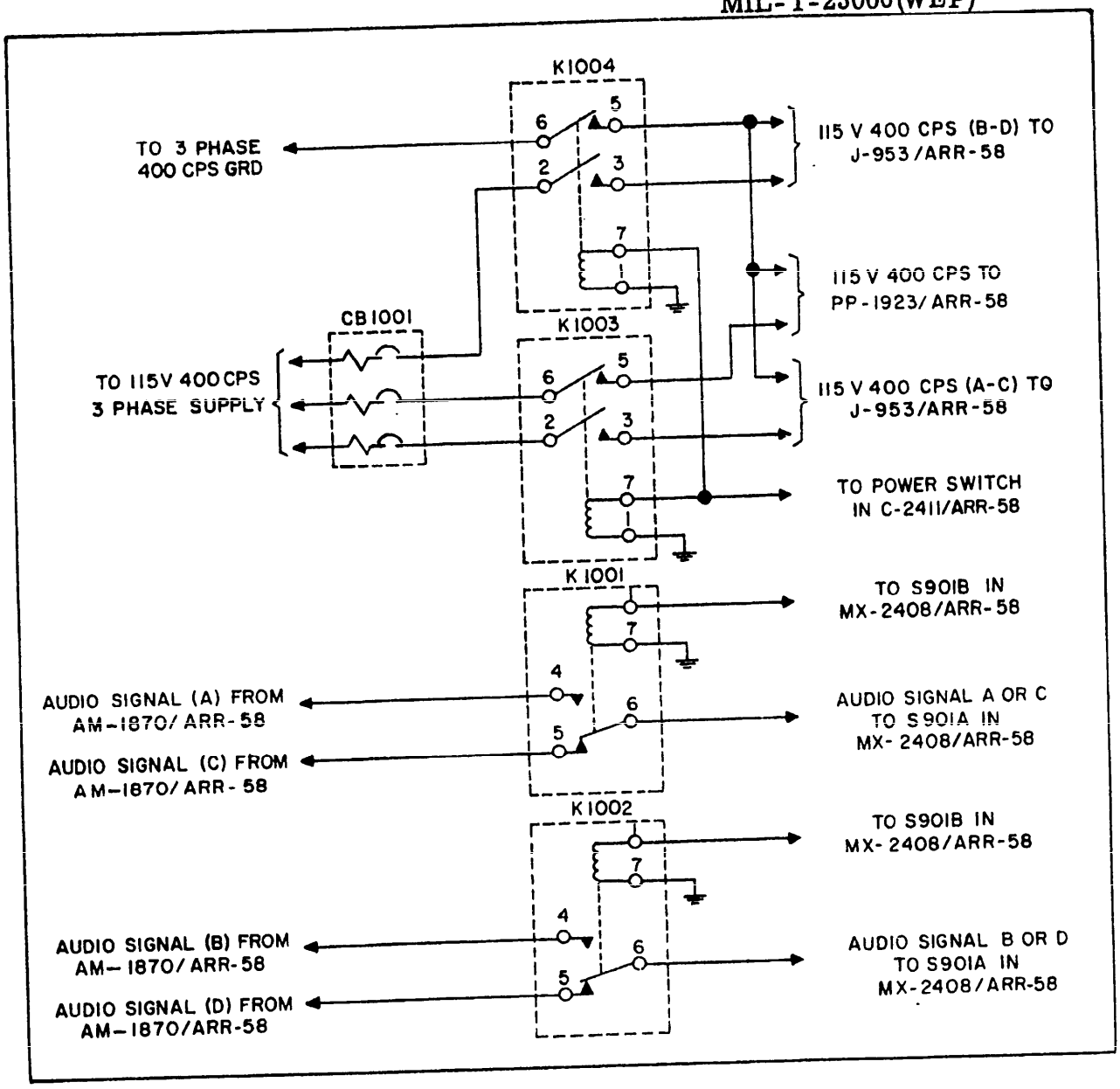



Figure 1. Switching Circuits of Interconnecting Box J-952/ARR-58



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(2) Power Supply Cable W-2. This cable assembly shall be approximately 7 feet in length and shall have four AWG #18 conductors, two of them shielded. The ends shall have the following plugs.

| P601 | P1002 |
|---|---|
| MS3106A-18-1S<br>MS3057-10A | MS3106A-18-1P<br>MS3057-10A |

(3) J953 and Low Audio Cable Assembly W-3. This cable assembly shall be approximately 9 feet in length and shall have two shielded AWG E18 conductors, eleven AWG #18 conductors, three AWG #20 conductors, and two RG-58C/U conductors. The ends shall have the following plugs.

| P506 (A-C) | P1001 |
|---|---|
| MS3108A-22-14S<br>MS3057-12A | MS3106-22-14P<br>MS3057-12A |
| P302A and P302C | P1003 |
| Magnavox No. 188235-2 | Magnavox No. 188235-4 |
| | P1006<br>Magnavox No. 188235-3 |



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(4) J-953 and Low Audio Cable Assembly W-4. This cable assembly shall be approximately 9 feet in length and shall have same type and number of conductors as in 3.4.6.2(3). The ends shall have the following plugs.

| P506 (B-D) | P1016 |
|---|---|
| MS3108A-22-14S<br>MS3057-12A | MS3106A-22-14PX<br>MS3057-12A |
| P302B and P302D | P1017 |
| Magnavox No. 188235-2 | Magnavox No. 188235-2 |
| | P1020 |
| | Magnavox No. 188235-1 |

(5) Monitor Panel Cable W-5. This cable assembly shall be approximately 8 feet in length and shall have two RG-188/U conductors. The ends shall have the following plugs.

| P507 (A-C) | P907 |
|---|---|
| MX #185070-2<br>(Dage #5143-1) | MX #185070-2<br>(Dage #5143-1) |
| P505 (A-C) | P906 |
| MX #185070-1<br>(Dage #1-670-1) | MX #185070-1<br>(Dage #1-670-1) |



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(6) Monitor Panel Cable W-6. This cable assembly shall be approximately 8 feet in length and shall have two RG-188/U conductors. The ends shall have the following plugs.

| P507 (B-D) | P909 |
|---|---|
| MX #185070-2<br>(Dage #5143-1) | MX #185070-4<br>(Dage #5146-1) |
| P505 (B-D) | P908 |
| MX #185070-1<br>(Dage #1-670-1) | MX #185070-3<br>(Dage #5052-1) |

(7) Monitor Panel Cable W-7. This cable assembly shall be approximately 5 feet in length and shall have six AWG #18 conductors and two RG-58C/U conductors. The ends shall have the following plugs.

| P902 | P1012 |
|---|---|
| MS3106A-10SL-4S<br>MS3057-4A | MS3106A-14S-5P<br>MS3057-6A |
| P901 | P1026 |
| MS3106A-12SL-844S<br>MS3057-6A | MS3106A-14S-5PX<br>MS3057-6A |
| P904 | P1023 |
| Magnavox No. 188235-3 | Magnavox No. 188235-4 |
| P905 | P1009 |
| Magnavox No. 188235-2 | Magnavox No. 188235-3 |



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(8) Control Set Cable (C-2411) W-8. This cable assembly shall be approximately 4 feet in length and shall have twelve AWG #18 conductors and three AWG #20 conductors. The ends shall have the following plugs.

| P702 | P1013 |
|---|---|
| MS3106A-24-28S | MS3106A-24-28P |
| MS3057-16A | MS3057-16A |

(9) Control Set Cable (C-2410) W-9. This cable assembly shall be approximately 3 feet in length and shall have eleven AWG #18 conductors and four AWG #20 conductors. The ends shall have the following plugs.

| P802 | P1027 |
|---|---|
| MS3106A-24-28S | MS3106A-24-28PX |
| MS3057-16A | MS3057-16A |

3.4.6.3 Function of Special Test Cable Assemblies - The Special Test Cable Assemblies shall allow operation of the R-863/ARR-58 Radio Receiver, AM-1870/ARR-58 Amplifier Audio Frequency, and AM-1871/ARR-58 Amplifier, Radio Frequency; when they are outside the J-953/ARR-58 Interconnecting Box-Power supply, where they are mounted during normal operation. Audio Output cables connect the AM-1870/ARR-58 Amplifier, Audio Frequency, properly loaded to and Electronic Multimeter.

3.4.6.4 Quantity and Type of Special Test Cable Assemblies - The Special Test Cable Assemblies shall consist of the following: (All wire shall meet the requirements of Specification MIL-E-16878 Type B.)

(1) Receiver Extension Cable Assembly W-10. This cable assembly shall be approximately 5 feet in length and shall have three AWG #18 conductors, seven AWG #20 conductors, three shielded conductors, and a twisted pair of AWG #22 conductors in a common shield. The ends of this cable assembly shall have the following plugs.

| P110 | P504 |
|---|---|
| Magnavox No. 188203-1 | Magnavox No. 188203-2 |



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(2) Amplifier Extension Cable Assembly W-11. This cable assembly shall be approximately 5 feet in length and shall have six AWG #22 conductors. The ends shall have the following connectors, Magnavox No. 185080-2 and 185080-1.

(3) High Level Audio Test Cable W-12. This cable assembly shall be approximately 6 feet in length and shall have an RG-58C/U conductor. One end shall have a Magnavox No. 188235-1 plug and the other end shall have a Magnavox No. 188326-1 plug with a 2400 ohm 1/2 watt resistor high level load.

(4) Low Level Audio Test Cable W-13. This cable assembly shall be approximately 6 feet in length and shall have an RG-58C/U conductor. One end shall have a Magnavox No. 188235-2 plug and the other end shall have a Magnavox No. 188326-1 plug with a 620 ohm 1/2 watt resistor low level load.

(5) Ground lead for bias supply W-14. This lead shall be approximately 3 feet in length and shall have one conductor and be terminated in an alligator clip and a terminal lug.

(6) RF Alignment test load W-15. This lead shall be approximately 5 feet in length and shall have an RG-62A/U conductor. One end shall be terminated with a Magnavox No. 188168-3 connector and the other end with a special contact holder having an RF isolation network consisting of a 68 ohm resistor and a 20 micromicrofarad capacitor.

(7) Bias Supply test lead W-16. This lead shall be approximately 3 feet in length and shall have one conductor terminated with a spade lug on one end and a special contact holder containing an RF coil on the other. *

* NOTE: A source of bias voltage is not supplied with Test Harness, Radio Receiving Set AN/ARM-44. Bias is to be obtained from a separate source such as a battery as shown in the alignment instruction portion of the AN/ARR-58 handbook.

3.4.7 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment.

| Item | Type |
|---|---|
| Table, Electrical Test and Maintenance | Bureau of Ships No. 5000-S9200-F1385763 |
| Radio Receiving Set | AN/ARR-58 |



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, including the Sand and Dust Test and modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181. Drop tests for transit and combination cases shall be in accordance with Specification MIL-T-21200.



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4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests -When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests



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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

| | |
|---|---|
| Temperature | Ambient room |
| Humidity | Ambient room |
| Vibration | Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3 g's |

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.



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4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

| Quantity of Equipments Offered for Acceptance | Quantity to be Selected For Sampling Test |
|---|---|
| First 10 | 1 |
| Next 50 | 1 |
| Next 75 | 1 |
| Next 100 | 1 |
| | 1 for each additional 200 or fraction thereof |

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)



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4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.



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## 5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

## 6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.



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6.4 The parentheses ( * ), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.9. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.



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